An Empirical Study of Compute Platform Influence on 5G gNB Performance and Energy Efficiency Using srsRAN

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***Abstract*—Lightweight 5G deployments for private networks and research testbeds demand platforms that balance performance, cost, and energy efficiency. While open-source stacks such as srsRAN and Open5GS have made 5G experimentation widely accessible, the impact of compute platform choice on gNB performance remains insufficiently characterized. This paper presents an empirical comparison of two heterogeneous hosts for an srsRAN gNB: a conventional x86 notebook and a Raspberry Pi. Using a controlled setup with a USRP B210 radio, an Open5GS core, and a commercial Nokia X30 UE, we evaluate downlink throughput via 20 repeated iPerf3 measurements alongside USB inline power meter profiling. Our results reveal a clear trade-off: the notebook completed all sessions reliably, whereas the Raspberry Pi suffered repeated USB disconnections arising from host controller instability under sustained high-bandwidth SDR transfers rather than from insufficient compute capacity. When sessions succeeded, the Pi achieved competitive throughput at approximately five times lower instantaneous power consumption. These findings suggest that ARM-based platforms are not fundamentally constrained for open-source 5G RAN workloads; rather, reliability gaps reflect software maturity and platform tuning, pointing to a concrete direction for future development.**

***Index Terms*—5G private network, energy efficient, sustainable 5G testbeds, performance evaluation**

I. Introduction

Open-source 5G software stacks such as srsRAN [1] and Open5GS [2] have significantly expanded the accessibility of 5G experimentation, enabling researchers, educators, and practitioners to deploy complete 5G Standalone (SA) networks using commodity hardware and SDR-based radio front-ends. In earlier work, we demonstrated the feasibility of constructing modular and cost efficient 5G SA networks entirely from open source components [3], and further showed that advanced services - including multimedia and voice - can be supported using Kamailio, a mature open-source SIP/IMS platform [4]–[6]. These capabilities are increasingly relevant for private networks and research testbeds, where support for VoNR, OTT voice, and mission-critical communication is becoming essential.

As lightweight and energy-efficient 5G deployments gain traction, especially in research labs, industrial testbeds, and small-scale private networks, the choice of compute platform becomes a critical design factor. While most open-source 5G deployments rely on x86-based systems, modern ARM platforms such as the Raspberry Pi 5 offer compelling advantages in terms of cost, power consumption, and thermal efficiency. These characteristics make ARM-based systems attractive for educational environments, where institutions must scale hands-on 5G experimentation to larger student cohorts while maintaining low per-node cost and energy footprint.

Despite this growing interest, the practical implications of running a full 5G gNB and core network on ARM platforms remain insufficiently understood. Existing open-source deployments implicitly assume x86 hosts, and little is known about how architectural differences influence SDR throughput, system stability, or energy efficiency under realistic workloads. To the best of our knowledge, no prior study has provided a controlled, empirical comparison of heterogeneous compute platforms running identical open-source 5G RAN and core stacks.

This paper addresses this gap by conducting a controlled empirical comparison of an x86 notebook and a Raspberry Pi 5 executing the same srsRAN gNB and Open5GS core, paired with a USRP B210 SDR and a commercial 5G UE. By measuring downlink throughput, instantaneous power consumption, thermal behavior, and system stability across repeated sessions, we quantify the trade-offs between performance, energy efficiency, and reliability. Our findings provide new insights into the feasibility of ARM-based platforms for lightweight 5G deployments and highlight concrete areas where software maturity and platform tuning can improve the robustness of open-source 5G RAN implementations.

The remainder of this paper is structured as follows: Section II reviews related work on open-source 5G RAN and core deployments, lightweight SDR-based testbeds, and prior evaluations of compute-platform constraints in software-defined 5G systems. Section III describes the experimental setup. Section IV presents the performance evaluation and comparative analysis. Section V discusses the broader implications of the results, while Section VI outlines

the key limitations of the study. Finally, Section VII concludes the paper and identifies directions for future research.

II. Related Work

Prior work has extensively benchmarked open-source 5G stacks on x86 commodity hardware, with most studies focusing on throughput, latency, and functional validation rather than compute-platform constraints. Alves et al. [7] deployed both srsRAN and OAI on Dell OptiPlex 3070 mini-PCs paired with USRP B210 radios and a Moto G200 UE, evaluating downlink throughput via iPerf and SpeedTest. Their results demonstrated that OAI achieved up to 123 Mbps at 40 MHz bandwidth compared to 62 Mbps for srsRAN, attributing the gap to the difference in OAI and srsRAN's support for downlink Modulation and Coding Scheme - a limitation since resolved in later srsRAN releases. Importantly, their study confirmed that both stacks are viable for full 5G SA experimentation with COTS UEs, but did not examine energy consumption or platform-level bottlenecks. Motaung et al. [8] conducted a broader comparative analysis of OAI and srsRAN, extending evaluation beyond throughput to encompass Docker integration, latency, and video streaming quality. Using a Dell Precision 3630 workstation with a USRP X310 and Open5GS, they found that OAI provides a more production-ready, modular containerised deployment, while srsRAN offers superior ease of use for beginners and higher uplink throughput at 13.4 Mbps versus OAI's 6.0 Mbps.

In addition to benchmarking-oriented studies, recent work has explored accessible 5G experimentation platforms based on open-source and cloud-native principles. Our previous work, OPEN6GNET [9], introduced a low-cost, educational platform designed to support hands-on learning and early-stage research by enabling students and newcomers to deploy and inspect 5G network components using containerised open-source tools. While OPEN6GNET focuses on pedagogy and reproducible experimentation rather than performance or energy evaluation, it highlights the growing need for lightweight, flexible 5G testbeds. The present study builds on this direction by shifting the emphasis from accessibility to compute-platform efficiency, examining how identical srsRAN deployments behave on different hardware architectures under real power profiling.

Across all these studies, two gaps remain unaddressed: energy consumption of 5G RAN workloads, and comparative evaluation of non-x86 architectures, particularly ARM.

Although these gaps persist within the 5G-RAN literature, research in adjacent domains has shown that ARM architectures can deliver superior power efficiency compared to x86 under a variety of compute workloads. Zhao et al. [10], for example, demonstrated that ARM-based servers can significantly reduce energy usage in data-center environments while maintaining competitive performance. Such findings motivate the inclusion of ARM platforms in emerging 5G research, especially as lightweight private 5G deployments increasingly target cost- and energy-constrained environments.

Building on these insights, the present work provides the first direct comparison of x86 and ARM platforms running identical srsRAN 5G SA deployments, with explicit power profiling under controlled conditions. By evaluating an Intel Core i7 notebook and a Raspberry Pi 5 (Cortex-A76), this study addresses the lack of architecture-aware energy analysis in prior open-source 5G evaluations.

III. Experimental Setup

The experimental testbed implements a complete software-defined 5G NR downlink chain built entirely from open-source components. Each host platform under evaluation runs an all-in-one 5G deployment consisting of the srsRAN gNB, a co-located Open5GS core network, and an iPerf3 client generating downlink traffic. A USRP B210 software defined radio (SDR), connected via USB 3.0, serves as the RF front-end, transmitting over-the-air to a commercial Nokia X30 handset acting as the COTS UE and iPerf3 server. This setup ensures that all traffic traverses the full 5G NR air interface rather than a simulated or bypassed link. The logical topology is shown in Fig. 1.

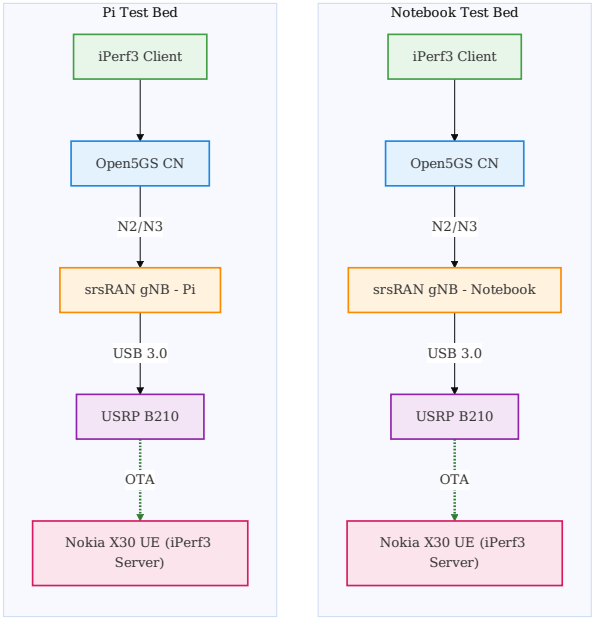


Fig. 1: Logical topology of the experimental setup. Each host platform operates the full 5G stack and connects to a shared USRP B210 SDR, which transmits OTA to a Nokia X30 UE.

Two host systems were evaluated to compare x86 and ARM compute architectures under identical workloads. The first is a Lenovo ThinkPad T14 notebook equipped with an Intel Core i7-10510U processor with 32GB RAM, which is a mobile CPU with four physical cores (eight logical threads via Hyper-Threading), a base clock of 1.8 GHz, and a boost frequency up to 4.9 GHz. The second is a Raspberry Pi 5 with 16GB RAM powered by a Broadcom BCM2712 integrating an ARM Cortex-A76 quad-core cluster clocked at up to 2.4 GHz without simultaneous multi-threading. Both platforms share

the same USRP B210 SDR unit, connected via USB 3.0, and operate Ubuntu 24.04 LTS. To ensure fairness, each platform executed the recommended srsRAN performance tuning script [11]. Table I summarizes the key architectural differences.

TABLE I: CPU Specifications: Intel Core i7-10510U vs ARM Cortex-A76

Attribute	Notebook	Pi
CPU	Intel Core i7-10510U	ARM Cortex-A76
Architecture	x86_64	AArch64
Physical cores	4	4
Logical threads	8 (Hyper-Threading)	4 (no SMT)
Base / Max clock (MHz)	1800 / 4900	1500 / 2400
L1d cache (per core)	32 KiB	64 KiB
L1i cache (per core)	32 KiB	64 KiB
L2 cache (per core)	256 KiB	512 KiB
L3 cache (total)	8 MiB	2 MiB
Estimated TDP	∼15 W	∼5 W
Approximate platform cost	€ 1500	€ 120

The USRP B210 connects to the host over USB 3.0 and supports sampling rates up to $61.44\,MS/s$ [12]. A 40 MHz cell requires the full $61.44\,MS/s$ rate, producing about $1.97\,Gbps$ ($61.44\,MS/s \times 32\,bits$) of sustained I/Q traffic on the USB 3.0 bus. The Raspberry Pi 5 already showed USB disconnections at $20\,MHz$, which uses $30.72\,MS/s$, so $40\,MHz$ was not feasible because it would double the sustained I/O load. The evaluation therefore used $20\,MHz$, the highest cell bandwidth at which both platforms could establish and maintain a connection through the B210's USB 3.0 interface.

Both platforms run an identical software stack to isolate hardware effects from software variability. The gNB is configured on the 5G New Radio (NR) band n78 at a centre frequency of $3.75\,GHz$, with a channel bandwidth of $20\,MHz$ and a subcarrier spacing of $30\,kHz$. Table II lists the deployed components.

TABLE II: Software Configuration

Component	Notebook	Raspberry Pi
Operating system	Ubuntu 24.04 LTS	Ubuntu 24.04 LTS
srsRAN version	*25.10*	*25.10*
Open5GS version	*2.7.7*	*2.7.7*
iPerf3 version	*3.16*	*3.16*
SDR driver (UHD)	*4.6.0*	*4.6.0*
SDR hardware	USRP B210	USRP B210
UE	Nokia X30	Nokia X30

Power consumption is measured using an RD Tech TC66C USB-C Power [13] Monitor and Charging Protocol Analyzer (0.005–30 V, 0–5 A, ±1 % accuracy), placed inline between the host and its power supply. This captures the total system power draw, including gNB processing, Open5GS core functions, iPerf3 traffic generation, and OS-level overhead. Idle power is recorded prior to each run (30 s on the Raspberry Pi, 15 s on the notebook) with no active UE connection, providing a baseline for load-relative analysis.

Each platform executes 20 sequential downlink iPerf3 sessions, each lasting 10 s. The iPerf3 client runs on the gNB host and transmits to the UE-based server, ensuring that all traffic traverses the full 5G NR stack and OTA link. Throughput is sampled at one-second intervals, yielding up to ten interval measurements per session. Sessions in which throughput drop to zero and do not recover within the 10 s window are classified as failed transmissions, reflecting instability in the RAN or host compute pipeline. Power measurements are collected continuously throughout each session, enabling correlation between throughput, stability, and instantaneous power draw. Because the core network and gNB are co-located, the measurements reflect the complete energy footprint of a lightweight private 5G deployment.

IV. Result Analysis

A. Throughput Evolution

Fig. 2 presents the mean downlink throughput sampled at one-second interval across all 20 successful iPerf3 sessions, together with ± 1 standard-deviation bands. During the first 5 seconds, both platforms exhibit broadly comparable throughput profiles: the notebook stabilises around 35–40 Mbps and the Raspberry Pi remains close behind at 32–38 Mbps, indicating comparable initial PHY-layer performance.

Beyond the sixth second, however, the behaviour of the two systems diverges. The Raspberry Pi exhibits a pronounced increase in variance, with several intervals showing short-lived drops or oscillations. This instability suggests that sustained high-rate USB 3.0 transfers from the USRP B210 increasingly stress the Raspberry Pi's host-controller and memory-subsystem bandwidth, leading to reduced temporal consistency. In contrast, the notebook maintains a tightly bounded standard-deviation throughout the full 10-second window, with no observable performance degradation, consistent with the larger performance reserve of its x86 platform during sustained operation.

B. Mean Throughput Per Run

Fig. 3 shows the distribution of per-run mean throughput across all successful sessions. The notebook achieved a median throughput of 35.0 Mbps, while the Raspberry Pi achieved 33.8 Mbps (also summarized in Tab. III). This difference of only 3.4% demonstrates a negligible margin, confirming that the ARM Cortex-A76 is not computationally bottlenecked by the srsRAN gNB workload: when USB transfers completed without interruption, the Raspberry Pi delivered throughput statistically comparable to the x86 host. The spread of the Raspberry Pi's distribution is noticeably wider, however, with greater inter-session variability consistent with the increased tail latency observed in the throughput evolution (Fig. 2).

C. Power Measurements

1) Idle Power Draw: Fig. 4 illustrates the power draw on both platforms under idle conditions, with no load from either srsRAN or the UE. The Pi drew a near-constant ≈2.5 W with a standard deviation below 0.01 W, reflecting the deterministic power envelope of the ARM SoC under low-activity steady

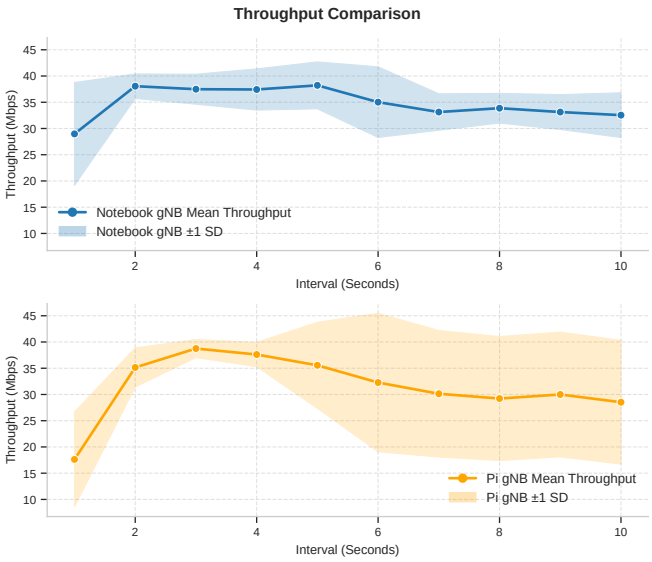


Fig. 2: Mean downlink throughput per one-second interval with ± 1 standard deviation over 20 successful sessions.

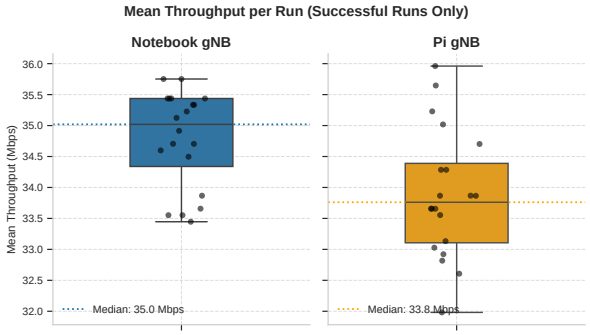


Fig. 3: Distribution of per-run mean throughput across all successful sessions. The notebook and Raspberry Pi differ by only 3.4%.

state. The notebook exhibited substantially higher idle power at ≈25 W with considerably greater variance, attributable to background OS scheduling, and display controller activity. This baseline difference of approximately 10× at idle already signals a significant energy efficiency advantage for the ARM platform before any RAN workload is applied.

2) Power Draw Under Load: Fig. 5 shows the full-stack power draw over time during the test campaign. The notebook sustained a stable ≈43 W throughout its 305 s run, completing

TABLE III: Stability and Throughput Variability Comparison

Metric	Notebook	Raspberry Pi
Sessions attempted	20	22
Successful sessions	20 (100%)	20 (90.9%)
Mean throughput (Mbps)	34.79	30.25
Median throughput (Mbps)	35.00	33.80
Standard deviation (Mbps)	0.78	9.82
Coefficient of variation (%)	2.2	32.5

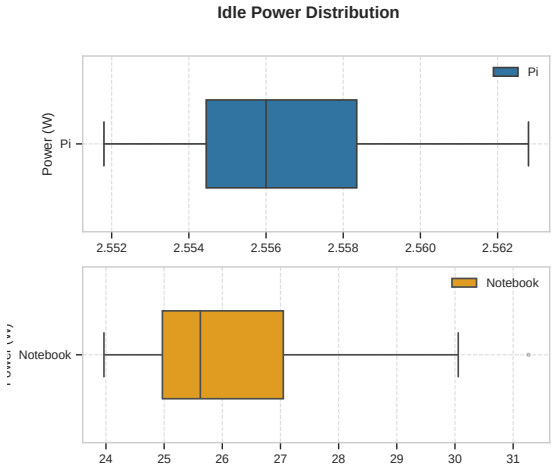


Fig. 4: Power draw distribution under idle conditions. The Pi draws a near-constant ≈2.5 W (std $<$ 0.01 W), while the notebook exhibits higher variability at ≈25 W, reflecting background OS scheduling activity.

all 20 sessions without interruption. The Pi exhibited a cyclic burst pattern at a median of 8.3 W, reflecting the alternating phases of active SDR transfer and brief idle recovery between iPerf3 sessions. Two USB disconnection events required manual restarts, extending the Pi's total experimental duration to 1478 s. Despite these interruptions, the Pi's median load power of 8.3 W represents a ≈5.2× reduction compared to the notebook's 43 W, underscoring the energy efficiency potential of the ARM platform for always-on 5G private network deployments.

Combining the median throughput from Section IV-B with the median load power yields an energy efficiency of $35.0 \div 43.0 = 0.81$ Mbps/W for the notebook and $33.8 \div 8.3 = 4.07$ Mbps/W for the Raspberry Pi, representing a 5.0× energy efficiency advantage for the ARM platform during successful transmissions.

3) Thermal Behaviour: Fig. 6 presents the CPU temperature over the duration of the test campaign under full-stack load. The notebook peaked at 43 °C and stabilised rapidly thereafter, benefiting from the active cooling system and larger thermal mass. The Pi remained consistently cooler throughout the experiment, peaking at 35 °C, well within the Pi's thermal limits. Notably, temperature dips are visible at both crash points on the Pi's trace, corresponding to reduced workload during USB disconnection and manual restart intervals. The absence of thermal throttling on either platform confirms that the observed throughput differences are attributable to USB host controller instability rather than thermal degradation of compute performance.

V. Discussion

The results highlight the practical trade-offs involved in deploying open-source 5G RAN stacks on low-power ARM platforms. While the Raspberry Pi 5 delivers throughput

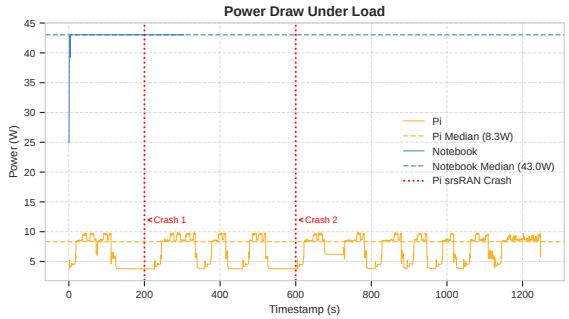


Fig. 5: Power draw over time under full-stack load (srsRAN gNB + Open5GS + iPerf3). The notebook sustains a stable ≈43 W throughout its 305 s run. The Pi exhibits a cyclic burst pattern at ≈8.3 W, interrupted by two USB disconnection events (LIBUSB_TRANSFER_NO_DEVICE) requiring manual restarts, extending the total session to 1478 s.

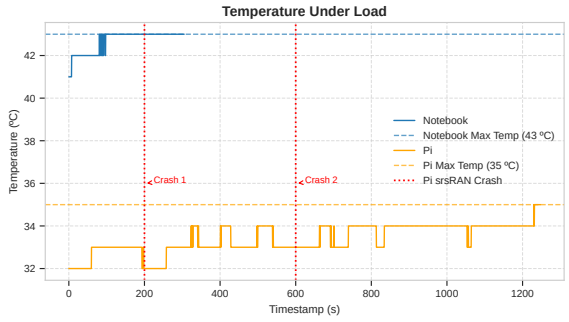


Fig. 6: Temperature under load. The notebook peaks at 43 °C and stabilises rapidly, while the Pi remains cooler throughout, peaking at 35 °C. Temperature dips at crash points reflect the reduced workload during USB disconnection and restart.

comparable to an x86 notebook when sessions complete successfully, its higher variance and occasional USB-related instability reveal the limits of lightweight hardware when sustaining continuous high-rate SDR traffic. At the same time, the Pi's 5× higher efficiency (4.07 Mbps/W vs. 0.81 Mbps/W) and stable thermal profile demonstrate that energy-efficient 5G deployments are feasible on compact ARM devices, provided that I/O subsystem reliability is addressed. These results suggest that ARM-based platforms, such as the Raspberry Pi 5, can support short or intermittent 5G deployments at bandwidths of up to $20\,MHz$. This makes them relevant for private 5G networks, edge testbeds, educational testbeds, periodic field trials, and cost-sensitive deployments where power consumption and form factor matter. However, continuous unattended operation would require either PCIe-attached SDR front ends to avoid the USB bottleneck or automated session recovery to restart sessions after transient disconnections. The results also point to the need for more robust USB and PCIe support in future low-power SDR platforms.

VI. Limitations

While the results demonstrate the feasibility of running a full 5G SA stack on low-power ARM hardware, the study has several limitations. First, all experiments were conducted with a single UE and downlink-focused traffic, which does not capture multi-UE contention or uplink scheduling behaviour. Second, the evaluation relies on the USRP B210, whose continuous high-rate USB 3.0 transfers expose platform-specific I/O bottlenecks; results may differ with PCIe-attached or lower-bandwidth SDRs. Third, the Raspberry Pi's instability stems from its USB and memory-subsystem constraints rather than compute limits, meaning that software-level optimisations alone cannot fully eliminate these effects. Finally, the study focuses on short-duration sessions and does not evaluate long-term reliability or performance under mobility. These constraints define the scope of the findings and motivate further investigation into more diverse hardware configurations and traffic scenarios.

VII. Conclusion & Future Work

This paper compared an x86 notebook and a Raspberry Pi 5 running an identical open-source 5G SA stack with a USRP B210 SDR. Both platforms achieved similar downlink throughput when sessions completed successfully, with median values differing by only 3.4%. The Raspberry Pi, however, showed higher short-term variance and occasional instability under sustained SDR load, reflecting the limited I/O headroom of its USB 3.0 and memory subsystem. In contrast, the notebook maintained stable performance throughout all runs.

Power measurements revealed a clear advantage for the ARM platform: the Raspberry Pi consumed roughly one-fifth of the notebook's load power while remaining thermally stable.

Overall, the results demonstrate that ARM-based platforms can deliver competitive 5G NR throughput at a fraction of the power budget, provided that their I/O subsystems remain stable. For lightweight private 5G networks, edge deployments, or educational testbeds, the Raspberry Pi offers a compelling energy-efficient alternative.

Future work will examine whether the USB issues can be avoided by moving the SDR off the USB bus. A likely approach is PCIe-attached SDR hardware, using the Raspberry Pi 5's PCIe interface where available. The next evaluation should also include uplink traffic and multi-UE tests with four to five simultaneous connections to measure how ARM-based platforms handle scheduling and resource allocation when several UEs compete for resources.

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[1] www.sustainet.de/sustainet-guardian/

References

[1] srsRAN GitHub Repository, "srsRAN: Open source O-RAN 5G CU/DU solution," https://github.com/srsran/srsran_project, Accessed: 2026-03-26.

[2] O. G. Repository, " Open5GS ," https://github.com/open5gs/open5gs, Accessed: 2026-03-26.

[3] M. Chepkoech, E.-R. Modroiu, J. Mwangama, M. Corici, and T. Magedanz, "Evaluation of oss-enabled openran compliant 5g standalone campus networks," in *2023 International Conference on Electrical, Computer and Energy Technologies (ICECET)*, 2023, pp. 1–7.

[4] E.-R. Modroiu, J. Mwangama, M. Corici, and T. Magedanz, "Integrating Open-Source VoNR in 5G SA Private Networks: An Experimental Evaluation," in *2024 International Conference on Software, Telecommunications and Computer Networks (SoftCOM)*, 2024, pp. 1–6.

[5] ——, "Experimental Evaluation of Open-Source-based VoNR and OTT Voice Services in 5G SA Network Deployments," *Journal of Communications Software and Systems*, vol. 21, no. 2, pp. 166–177, 5 2025. [Online]. Available: https://doi.org/https://doi.org/10.24138/jcomss-2025-0007

[6] K. G. Repository, "Kamailio: The Open Source SIP Server for large VoIP and real-time communication platforms," https://github.com/kamailio, Accessed: 2024-05-26.

[7] R. P. Alves, J. G. A. d. S. Alves, M. R. Camelo, W. O. de Feitosa, V. F. Monteiro, and F. R. P. Cavalcanti, "Experimental Comparison of 5G SDR Platforms: srsRAN x OpenAirInterface," in *XLI Brazilian Symposium on Telecommunications and Signal Processing (SBrT)*, São José dos Campos, SP, Brazil, Oct. 2023. [Online]. Available: https://arxiv.org/abs/2406.01485

[8] M. Motaung, N. Mthethwa, B. Nokane, and M. Masonta, "Comparative Analysis of 5G SDR-Based Open-Source Software Platforms: OpenAirInterface and srsRAN," in *2025 IEEE 3rd Wireless Africa Conference (WAC)*. IEEE, Feb. 2025, pp. 1–7.

[9] M. S. De Brito, E.-R. Modroiu, B. Q. Le, T. Magedanz, M. Corici, and J. Mwangama, "Open6gnet: A learning and experimentation platform based on open-source solutions and cloud-native approaches," in *IEEE INFOCOM 2024 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2024, pp. 1–6.

[10] e. a. Zhao, "Towards green data centers: A comparison of x86 and arm architectures power efficiency," *Journal of Parallel and Distributed Computing*, 2012. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S0743731512002122

[11] srsRAN Project, "srsran_performance," https://github.com/srsran/srsRAN_Project/blob/main/scripts/srsran_performance, 2023, Accessed: April 8, 2026.

[12] Ettus Research, "USRP B200/B210 Product Specification," https://www.ettus.com/wp-content/uploads/2019/01/b200-b210_spec_sheet.pdf, 2019, accessed: 2026-06-29.

[13] R. Tech, "Tc66c USB-C power monitor and charging protocol analyzer," https://lygte-info.dk/review/USBmeter%20RD%20Tech%20USB%20Meter%20TC66C%20UK.html, 2019, voltage: 0.005–30 V, Current: 0–5 A, Power: 0–150 W, Accuracy: ±(1% + 30 digits) @ 23°C.